

Youth Unemployment in Developing Countries: An Exploratory Data Analysis

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Abstract—This study presents an exploratory data analysis of youth unemployment in developing countries using data from the World Development Indicators (WDI) covering the period 2010–2022. The analysis examines the relationship between education levels, government expenditure on education, and youth unemployment rates across 177 developing countries. Through comprehensive data pre-processing and visualization techniques, we identify key patterns including regional disparities, the paradox of education, and the critical link between unemployment and NEET (Not in Education, Employment, or Training) rates. The findings reveal that youth unemployment disproportionately affects young people compared to the general population, with significant regional concentrations in African nations. The analysis also highlights a skills mismatch phenomenon where advanced education does not guarantee employment in many developing economies.

Index Terms—IEEE, IEEEtran, journal, L^AT_EX, paper, template.

I. INTRODUCTION

Youth unemployment refers to young people aged 15–24 who are actively seeking employment but are unable to find jobs. In developing countries, this issue is particularly acute due to a large youth population entering the labor market annually. Globally, more than 64 million young people are unemployed, with the majority residing in developing nations where 90% of youth live [1].

Youth unemployment has serious economic and social consequences:

- Lower consumer spending and tax revenues, increased fiscal burdens
- Poverty, psychological distress, crime, and potential political instability
- Brain drain as skilled youth emigrate for better opportunities

Addressing youth unemployment is crucial for economic growth, innovation, and social stability.

A. Project Description

1) *Goal/Purpose*: The primary goal of this analysis is to understand the determinants of youth unemployment in developing countries and identify patterns that may help policymakers improve employment opportunities for young people.

2) *Key Questions to be Answered*:

- 1) How do youth unemployment rates vary across countries and regions?

- 2) What is the relationship between education levels and youth unemployment?
- 3) How does government expenditure on education correlate with youth employment?
- 4) Are there trends or patterns over the period 2010–2022 that indicate improvement or worsening of youth unemployment?
- 5) What insights can be drawn specifically from the case study of Pakistan?

3) *Assumptions/Methodology/Scope*: **Exploratory Data Analysis (EDA)**:

- Examine distributions, correlations, and missing values

Data Preprocessing:

- Fix column names, data types, and categorical variables
- Handle missing values with linear interpolation and medians
- Drop duplicates and irrelevant or redundant columns

Assumptions:

- Education and government expenditure are proxies for youth skill development
- GDP per capita is a proxy for economic opportunities
- Trends in the dataset reflect broader national patterns

B. Data Description

1) *Content*: This dataset is an Excel file sourced from the *World Development Indicators (WDI)* and covers 2010–2022 for developing countries across Africa, Asia, Latin America, the Middle East, and North Africa. The dataset contains 2,665 rows and 67 columns initially, covering 177 countries after preprocessing.

2) *Acknowledgements*: This dataset is downloaded from the *World Development Indicators (WDI)* website and the original source can be found at <https://databank.worldbank.org/source/world-development-indicators>.

II. LITERATURE REVIEW

III. METHODOLOGY

A. Data Acquisition and Loading

The dataset was loaded from an Excel file containing World Development Indicators data. Initial exploration revealed:

- 2,665 rows and 67 columns
- Time period: 2010–2022
- Coverage: Developing countries across multiple regions

B. Data Preprocessing

1) Column Management: **Dropping Redundant Columns:**

- Gender-specific columns (female/male breakdowns) were removed to focus on aggregate data
- Time Code column was dropped as redundant

Renaming Columns:

- Columns were converted to snake_case format using the `clean_columns` function
- Long descriptive names were shortened to more manageable variable names (e.g., `unemp_youth` for youth unemployment)

2) Data Type Conversion:

- Country names and codes were converted to categorical data types
- Year was converted to integer type
- All numeric columns were converted to float type, handling errors with coercion

3) Handling Duplicates:

- Duplicate rows based on country and year combinations were identified and removed
- Only the first occurrence was kept for each unique country-year pair
- Verification confirmed zero remaining duplicates after processing

4) *Missing Value Treatment:* A comprehensive multi-stage approach was employed:

- 1) **Replacement of Placeholders:** Special values like ‘..’, ‘...’, ‘N/A’, ‘na’, ‘NA’, and empty strings were replaced with NaN
- 2) **Forward/Backward Fill:** Missing values were filled within each country using forward fill followed by backward fill
- 3) **Linear Interpolation:** Missing values were interpolated linearly within each country’s time series
- 4) **Country Median Fill:** Remaining missing values were filled with country-specific medians
- 5) **Global Median Fill:** For countries with no data, global medians were used as a last resort

After this pipeline, all numeric columns achieved 0% missing values.

5) Country Name Standardization:

- Country names were standardized using a mapping dictionary for common variations
- Regional aggregates and income groups were removed
- Territories and non-sovereign entities were excluded
- Final dataset contained 177 valid countries

6) *Feature Scaling:* All numeric features were standardized using Z-score normalization (StandardScaler) to ensure comparability across different scales of measurement.

C. Data Analysis and Visualization

The analysis employed multiple visualization techniques to explore patterns:

- **Distribution Analysis:** Violin plots and box plots to compare unemployment distributions

- **Correlation Analysis:** Arc diagrams and heatmaps to identify relationships between variables
- **Geographic Analysis:** Radial bar charts and choropleth maps to visualize regional patterns
- **Multivariate Analysis:** 3D scatter plots, bubble charts, radar charts, and treemaps for complex relationships

IV. RESULTS

A. Data Overview

After preprocessing, the final dataset contained:

- 177 countries
- Time period: 2010–2022 (13 years)
- All missing values successfully imputed
- Zero duplicate records

B. Key Findings from Visualizations

1) *Severity Comparison: Youth vs. General Unemployment vs. NEET:* Analysis of distribution densities revealed:

- **Disproportionate Impact on Youth:** Total unemployment shows a “short and wide” distribution, indicating relatively low and stable unemployment for the general population. In contrast, youth unemployment displays a “tall and stretched” distribution with significantly higher medians, revealing that youth face a much more volatile labor market. In many nations, youth are 2x or 3x more likely to be unemployed than adults.
- **The NEET Crisis:** The NEET (Not in Education, Employment, or Training) distribution shows patterns similar to youth unemployment but often with a wider base. This quantifies the “lost generation”—youth who are economically inactive and at risk of long-term poverty.

2) *Geographical Hotspots: Top 10 Countries by Youth Unemployment:* The analysis identified the top 10 countries with the highest average youth unemployment rates from 2010–2022:

- **Regional Concentration:** A strong cluster of African nations appears in the top 10 (e.g., Djibouti, South Africa, Eswatini, Libya, Congo, Namibia, Botswana). This suggests that youth unemployment is not just a national issue but a **structural regional crisis** in Africa, likely driven by a “youth bulge” demographics where the population is growing faster than job creation.
- **Magnitude of Disparity:** Djibouti leads with the highest rate. The visual gap between the outer rings and inner rings highlights that even among the “worst performers,” there is a significant difference in severity, requiring tailored intervention strategies for the most extreme cases.

3) *The Paradox of Education: Unemployment by Education Level:* Box plot analysis revealed a critical finding:

- **The “Skills Mismatch” Phenomenon:** Ideally, the “Advanced Education” category should show much lower unemployment than “Basic Education.” However, the chart shows significant overlap between groups. The presence of numerous outliers above the top whisker in the Advanced Education category is alarming, indicating

that in many developing countries, university graduates face exceptionally high unemployment rates.

- **Policy Implication:** This supports the theory that the economy is not creating enough white-collar/high-skill jobs to absorb the supply of graduates. Simply increasing university enrollment without fixing labor market demand may lead to a surplus of "educated unemployed" youth.

4) *Variable Relationships: Correlation Analysis:* Arc diagram and heatmap analysis revealed:

- **The NEET-Unemployment Link:** A strong connection between youth unemployment and NEET rates confirms that as unemployment rises, the percentage of youth "Not in Education, Employment, or Training" also rises. This signals a risk of social exclusion; unemployed youth are likely to drop out of the system entirely rather than returning to education.
- **Education Expenditure & Outcomes:** The relationship between Government Expenditure (as % of GDP or on tertiary education) and outcome variables is critical for policy analysis. Weak connections may indicate fiscal inefficiency—where money is being spent on education but is not effectively translating into jobs for graduates.

C. Multivariate Analysis Results

1) *3D Scatter Plot: Government Spending vs Education vs Youth Unemployment:* The three-dimensional visualization exploring government education spending, bachelor's attainment, and youth unemployment revealed complex relationships that vary by country and over time.

2) *Country Comparison: Radar Chart:* Multi-dimensional comparison of Pakistan, India, Bangladesh, Brazil, and South Africa across six key variables (youth unemployment, government expenditure on education, bachelor's attainment, tertiary expenditure, advanced education labor force, and NEET rates) showed distinct profiles for each country, highlighting the need for country-specific policy approaches.

V. DISCUSSION

A. Interpretation of Key Findings

1) *Regional Disparities:* The concentration of high youth unemployment rates in African nations suggests structural issues beyond individual country policies. The "youth bulge" phenomenon, where a large proportion of the population is young, creates pressure on labor markets that cannot be addressed through education alone. This requires comprehensive economic development strategies that create jobs while simultaneously improving education quality.

2) *The Education Paradox:* The finding that advanced education does not guarantee lower unemployment challenges conventional wisdom. This "skills mismatch" indicates a fundamental disconnect between educational output and labor market demand. Possible explanations include:

- Overproduction of graduates in fields with limited job opportunities
- Mismatch between skills taught and skills demanded by employers

- Insufficient job creation in knowledge-intensive sectors
- Brain drain of highly educated individuals to developed countries

3) *The NEET Crisis:* The strong correlation between youth unemployment and NEET rates is particularly concerning. When young people become unemployed, they often disengage from both education and the labor market entirely. This creates a "lost generation" that may never fully integrate into the economy, leading to long-term social and economic costs.

4) *Policy Implications:* The weak correlation between government education expenditure and employment outcomes in some cases suggests that:

- **Spending alone is insufficient:** The quality and relevance of education matter more than the quantity of spending
- **Coordination is needed:** Education policy must be coordinated with economic and labor market policies
- **Targeted interventions:** Different countries require different approaches based on their specific economic structures and labor market conditions

B. Limitations

- The analysis relies on aggregate country-level data, which may mask subnational variations
- Missing value imputation, while comprehensive, may introduce some uncertainty
- The time period (2010–2022) may not capture longer-term trends
- Correlation does not imply causation; further econometric analysis would be needed to establish causal relationships

VI. CONCLUSION

This exploratory data analysis of youth unemployment in developing countries has revealed several critical insights:

A. Key Findings

- 1) **Youth face disproportionate unemployment:** Young people are 2–3 times more likely to be unemployed than the general population, indicating a systemic issue affecting this demographic group.
- 2) **Regional concentration in Africa:** The highest youth unemployment rates are concentrated in African nations, suggesting structural regional challenges that require coordinated regional responses.
- 3) **Education paradox:** Advanced education does not guarantee employment, revealing a skills mismatch between educational output and labor market demand.
- 4) **NEET crisis:** High unemployment correlates strongly with disengagement from both education and employment, creating risks of long-term economic exclusion.
- 5) **Policy inefficiency:** Government education expenditure does not always translate into improved employment outcomes, indicating the need for better coordination between education and economic policies.

B. Policy Recommendations

Based on these findings, we recommend:

- **Job creation focus:** Policies should prioritize creating employment opportunities, particularly in sectors that can absorb educated youth
- **Education quality and relevance:** Reform education systems to align curricula with labor market demands and improve quality rather than just increasing enrollment
- **Regional cooperation:** African nations should coordinate youth employment strategies given the regional nature of the crisis
- **Targeted interventions:** Develop country-specific strategies based on local economic structures and labor market conditions
- **NEET prevention:** Implement programs to prevent unemployed youth from disengaging entirely from education and training systems

C. Future Research Directions

- Conduct econometric analysis to establish causal relationships between education expenditure and employment outcomes
- Analyze subnational data to identify within-country variations
- Investigate the role of specific economic sectors in youth employment
- Examine the effectiveness of different policy interventions across countries
- Study the long-term impacts of youth unemployment on economic development

Addressing youth unemployment is crucial for economic growth, innovation, and social stability in developing countries. This analysis provides a foundation for evidence-based policy development, though further research is needed to develop and evaluate specific interventions.

REFERENCES

- [1] World Development Indicators (WDI), 2024. World Bank DataBank. Available at: <https://databank.worldbank.org/source/world-development-indicators>